fit@hcmus

Object-Oriented Programming

Course Introduction



Tran Duy Hoang

Teaching Staffs

- Instructor
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- Teaching assistant
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Course Description

- Provides students
  - object-oriented concepts
  - problem analysis, design and solving
  - advanced C++ concepts and features

- Students could
  - describe key object-oriented concepts
  - solve problems using object-oriented programming

Topics Covered

- Function overloading, template, general programming
- Introduction to object-oriented programming
- Constructors and destructor
- Operator overloading
- Standard template library
- Inheritance
- Polymorphism
- Design patterns
- Exception handling

Text books

- Required
  - ***Phương pháp lập trình hướng đối tượng***, Trần Đan Thư, Đinh Bá Tiến, Nguyễn Tấn Trần Minh Khang, Nhà xuất bản Khoa học và Kỹ thuật, 2020
  - ***C++ Primer Plus***, Stephen Prata, 6th edition, Addison-Wesley, 2011

- Optional
  - ***Design Patterns: Elements of Reusable Objected-Oriented Software***, Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, Addison-Wesley, 1995
  - ***Effective C++: 55 Specific Ways to Improve Your Programs and Designs***, Scott Meyers, Addison-Wesley, 3rd edition, 2005
  - ***More Effective C++: 35 New Ways to Improve Your Programs and Designs***, Scott Meyers, Addison-Wesley, 2nd edition, 1996
  - ***The C++ Programming Language***, Bjarne Stroustroup, 4th edition, 2013

Course Requirements

- Moodle used for material distribution and communication
- Questions beneficial to both the questioner and others should be posted on Moodle's forum
- Students encouraged to ask questions in class, via forum, email, or in-person
- Late submissions will not be accepted

Grading

- Grade Distribution
  - Lab grade 30%
  - Theory grade 70%
    - Assignments 10%
    - Mid-term exam 10%
    - Final exam 50%

Academic integrity

- Students are prohibited from copying
  - from classmates, friends even if allowed
  - from the Internet without proper citation
- Students are prohibited from allowing others to copy
- Other kinds of cheating and plagiarizing

- If the academic integrity violated, serious measures will be taken
  - 1st violation: zero grade for the assignment violating
  - 2nd violation: students will be failed the class and report to the Faculty